

Wonbeom Lee

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RESEARCH INTERESTS

Systems for AI, Computer Architecture, Hardware-Software Co-Design

EDUCATION

M.S./Ph.D. in Electrical and Computer Engineering

03/2023-Present

Seoul National University

Computer Architecture and Systems Lab (Advisor: Prof. Jaewoong Sim)

B.S. in Electrical and Computer Engineering

03/2019-08/2022

Seoul National University

Early Graduation, GPA: 3.84/4.30, major GPA: 3.94/4.30

PUBLICATIONS

[OSDI '24] **InfiniGen: Efficient Generative Inference of Large Language Models with Dynamic KV Cache Management**

Wonbeom Lee*, Jungi Lee*, Junghwan Seo, Jaewoong Sim

Acceptance Rate: 49/282 $\approx$ 17%

[ISCA '24] **Tender: Accelerating Large Language Models via Tensor Decomposition and Runtime Requantization**

Jungi Lee*, Wonbeom Lee*, Jaewoong Sim

Acceptance Rate: 83/423 $\approx$ 19%

RESEARCH EXPERIENCES

Research Assistant

03/2023-Present

Seoul National University (Advisor: Prof. Jaewoong Sim)

- **Developed a KV cache management framework for efficient long-context LLM serving.**
 - Proposed **InfiniGen**, a novel KV cache management framework tailored for long-text generation that synergistically works with modern offloading-based inference systems.
 - Designed a minimal rehearsal algorithm to prefetch a few important tokens that are essential for computing the subsequent layer, thereby mitigating the data transfer overhead in offloading-based LLM serving systems.
 - Implemented InfiniGen and baselines on the FlexGen framework, achieving up to 3 $\times$ speedup.
- **Worked on algorithm-hardware co-design for accelerating LLM inference.**
 - Designed **Tender**, an algorithm-hardware co-design solution that offers high performance and accuracy without the need for mixed-precision compute units or custom data types even for low-bit quantization.
 - Proposed a decomposed quantization technique in which the scale factors of the decomposed matrices are powers of two apart, enabling implicit requantization with negligible rescaling overhead and minimal hardware extension.
 - Observed the value distribution of LLM activations and implemented a simulator for Tender accelerator using C++.

TEACHING EXPERIENCES

Seoul National University

Spring 2023

Graduate Teaching Assistant

• **430.322: Computer Organization**

Designed part of course projects, led recitation sessions, and graded projects/exams

SKILLS

- **Languages:** C/C++, Python, Triton, CUDA
- **Applications/Frameworks/Tools:** PyTorch, vLLM, SGLang, Intel Pin, Nsight Systems